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ICS 2203 WEB APPLICATION DEVELOPMENT 1

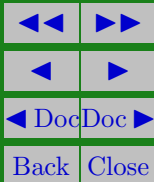
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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.





LESSON 9

Introduction to CSS (cascaded Style Sheets)

Learning outcomes

Upon completing this topic, you should be able to:

1. Identify CSS web technologies in use in web programming.
2. Employ CSS technology in styling web pages .
3. Generate web pages though the use XHTML and CSS.

9.1. Introduction to CSS (cascaded Style Sheets)

In the advent of HTML it was designed to specify the syntax for laying out a web page. This Intent however was lost after a short



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while when many authors discovered new ways of writing tags that affected the look and feel of the web page. Tags such as the font tag with its many attributes were being used to format web documents and not really controlling the layout of the different sections that make up the web pages.

HTML version 4 was the climax of this and this lead to web authors spending more time formatting web pages than concentrating on the layout. This in time complicated web programming and the World Wide Web consortium in response to this crisis came up with Style Sheets to deal with the look and feel while HTML to deal with the layout.



9.1.1. What is a Style Sheet?

These are special scripts that specify rules for styling web pages. They are usually defined within the style tag in HTML /XHTML

- A Cascading Style Sheet is a file with a list of formatting instructions.
- CSS style sheets are the modern way to control the appearance and layout of your web pages.

Every element on a page has a style defined for it whether custom or default. Style is defined by a set of attribute : value pairs.

- Typeface and font properties i.e. font type, font color, font family e.t.c
- Background properties i.e. background color, background graphics e.t.c



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- Box-related properties i.e. table styles
- List properties i.e. numbered or bulleted lists
- **Ways of Defining Style**
 - Default style: as earlier stated each element has a default style provides values for all element properties. Unless this changes the default style is applied throughout the page.
 - Inline style: style is defined as an attribute of the element in-place. It is used for “one-off” specification for special styles. In many cases the style rule is not re used.
 - External style sheet: here styles defined in a separate style sheet file which has a .css file name extension. It is used to centralize style definitions and provide uniformity across all pages of a web site.



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- Embedded style sheet: are styles defined in the head portion of web page. It is preferred if you don't have very many web pages, or for styles that are useful only for the given page.

Example of an embedded style

```
<HTML>
<head>
<title>Page with embedded style</title>
<style type="text/css">
selector{attribute : value ;
attribute :value...}
selector{attribute : value ;
attribute :value...} ...
</style>
```



```
</head> ...
```

```
</HTML>
```

- **Style Sheet Rules**

These are defined following the syntax

```
selector{attribute : value ; attribute2 :value2...;}
```

Example . example of a style sheet that will affect the heading style and the paragraph style. it creates a centered heading with a type font of arial color blue , while a paragraph will be left aligned, color will be red and font family will be tahoma. in addition the text will be italicized H1 {text-align: center; color: blue; font: Arial}

P {text-align: left; color: red; font-family: Tahoma, Arial Narrow; font-style: italics}



```
<html>
<head>
<title>example page layout using table</title>
</head>
<body>
<table width="100%"height="186"style=
"border-collapse:collapse" border="1">
<tr>
<td width="14%" height="97"><imgsrc="atinga.jpg"
width="91" height="89" alt="our logo" /></td>
<td colspan="2"><div align="center">
<strong>Heading goes here</strong></div></td>
</tr>
```



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What is a selector? A selector may be an element in HTML or a class where the CSS rules will be applied. In the case of elements we have simple selector like the ones in the previous examples and compound selectors where a rule is specified for a number of elements. i.e. **ulul { list-style-type : square }** This specifies that an unordered list inside another unordered list will be bulleted by squares. **h1emem { color : red }** This specifies that emphasized text inside emphasized text in an `<h1>` header will appear in red.

class selectors

These allow you to control which elements of a given type should use a style rule. This method has two parts:

- In the style sheet, the selector defines the class name, which is preceded by a period.



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- In the HTML, the tag includes the class attribute and specifies the value of the class name

Example . Define the lyx class for anchor tags:

- `a.lyx { text-decoration : none }`

This suppresses the usual underlining. Use it in HTML like so:

```
<a class="lyx" href="somepage.html">Link text</a>
```

- **Generic style classes**

There are instances where we want to define a class style that is not tied to a specific element. In this case we can use style classes that are generic i.e. not tied to a specific element type.

Example:

Define the bling class:

- `.bling { text-decoration : blink }`



Use it on an emphasized element:

```
<em class="bling">Important!</em>
```

Use it with no other style attributes:

```
<span class="bling">Buy Now!</span>
```

The span and div elements

- These tags are provided to allow arbitrary chunks of HTML to be treated as elements. This is usually done in order to apply style attributes to them, as in the preceding example
- A `<span> ... </span>` element defines an “inline” structure, i.e. it simply defines a stretch of text. Thus it can be used within a paragraph or table element without affecting the flow of the text.
- A `<div> ... </div>` element defines a “block” structure. Usually the browser will place line breaks before and after

this element, but otherwise it has no effect itself.

The cascading rule

- If two equally specific rules can apply to the same element, the one that comes last in the style sheet is used. Thus, in the example below, **a:hover** must follow **a:link** and **a:visited** in order to have effect, since a link can be both visited (or not) and hovering. The order of the first two doesn't matter since they are mutually exclusive.
- **a:link { color : red }**
- **a:visited { color : green }**
- **a:hover { color : yellow }**





- **Inline Style**

- Defined for individual elements, at the point of use (in the HTML).
- Useful for “one-off” styles that are not likely to be used elsewhere.
- Method:

`<tag style="attribute:value;attribute:value...">HTML text</tag>`

The attribute:value pairs are what would go between { } if this were a style-sheet rule. There is no selector since the style applies to this element only.

- **External style sheets**

- A style sheet can be placed in a separate file (usually named with suffix .css) and referenced by HTML files that



need it.

- Useful for centralizing control of style and ensuring uniform appearance across many pages of a web site.
- The contents of the file are what would go between `<style>` ... `</style>` of an embedded style sheet. Note: this file is not HTML!
- The file is referenced using a `<link>` tag in the HTML document's head portion.

Here is what an external style sheet (named style.css) could contain:

```
}  
p.exstyle{ font-style : italic;  
margin-right : 2em;
```



```
}
```

Note that there is no HTML in this file! This example also illustrates a CSS comment between `/*` and `*/`. Such comments can be placed in external or embedded style sheets

- Using an external style sheet in html

```
<HTML>
```

```
<head>
```

```
<title>Style Example</title>
```

```
<link rel="stylesheet" type="text/css" href="style.css">
```

```
</head>
```

```
... <p class="exstyle">affected text</p>...
```

```
</HTML>
```

- The rel attribute specifies the relationship of the referenced



file to this page.

- The type attribute must be "text/css".
- The href attribute is a URL pointing to the external style sheet.

Example . Describe the syntax of a well formed XML document (3 marks).

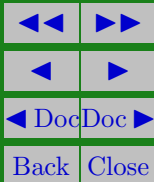
Solution: A well-formed document conforms to the XML syntax rules; e.g. if a start-tag (< >) appears without a corresponding end-tag (</>), it is not well-formed. A document not well-formed is not in XML; a conforming parser is disallowed from processing it. Valid. A valid document additionally conforms to semantic rules, either user-defined or in an XML schema, especially DTD; e.g. if a document contains an undefined element,

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then it is not valid; a validating parser is disallowed from processing it. i.e. `<book>This is a book... </book>`



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Revision Questions

EXERCISE 1. 🖐 Define a style sheet

EXERCISE 2. 🖐 Define a star class for the anchor tag that specifies a comic sans font type with a size of 32

EXERCISE 3. 🖐 Describe three ways of specifying style sheets

EXERCISE 4. 🖐 Illustrate how you would link an external stylesheet known as armload. css to a html file

References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp



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- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5



Solutions to Exercises

Exercise 1.

This is a rule written in a scripting language on how the look and feel aspects of a web page will be controlled . **Exercise 1**



Exercise 2.

a. star (2mks) {font-type: comic sans; font-size: 32;}

Exercise 2



Exercise 3.

inline style sheets

embedded style sheets References and Additional Reading

Materials

external style sheets

default style sheets

Exercise 3